Léo Guignard, Ph.D.

CURRENT POSITION

CNRS Chargé de Recherche 2024 -
Institut de Biologie du Développement de Marseille
How are different developmental variabilities influencing embryogenesis?

Group Leader - "Computer Science, Morphogenesis and Variability" 2023 -
Institut de Biologie du Développement de Marseille – Turing Centre for living systems, Marseille, France
How are different developmental variabilities influencing embryogenesis?

RESEARCH EXPERIENCE

Visiting Scientist (Funding: Humboldt renewed research stay) 2023 – 2024
Max Delbrueck Centre for Molecular Medicine, Berlin, Germany
AI methods to automatically reconstruct mouse embryo nervous system from fluorescence imaging.

Group Leader 2021 – 2023
Laboratoire d'Infomatique et Systèmes – Turing Centre for living systems, Marseille, France
How are different developmental variabilities influencing embryogenesis?

Postdoctoral Associate 2019 – 2021
D. Kainmueller group, Max-Delbrueck Centre, Berlin, Germany
Reconstruction of average *Drosophila* embryonic development as spatio-temporal graphs and their quantitative analysis

Postdoctoral Associate 2016 – 2019
P. Keller group, Janelia Research Campus, Ashburn, VA, USA
Reconstruction of average Mouse embryonic development as spatio-temporal graphs and their quantitative analysis

Master thesis internship 02/11 – 06/11
Under the supervision of G. Melançon, LaBRI, Bordeaux, France
Dynamic graph clustering

Undergraduate internship 04/08 – 06/08
Under the supervision of D. Dupuy, IECB, Bordeaux, France
Development of custom software to parse, sort and analyze raw data from high throughput biological studies of gene expression

WORK EXPERIENCE

Software developer 2008 – 2011
IECB, Bordeaux, France
Development of custom software to parse, sort and analyze raw data from high throughput biological studies of gene expression

EDUCATION

Ph.D. in Biology 2011 – 2015

Supervised by P. Lemaire (CRBM) and C. Godin (Inria), Montpellier, France

Quantitative analysis of animal morphogenesis: from high-throughput laser imaging to 4D virtual embryo in ascidians.

Master's in Computer Science 2009 – 2011

University Bordeaux 1, France

Algorithmic and Formal Methods

Bachelor's in Computer Science 2006 – 2009

University Bordeaux 1, France

FUNDINGS

Name: **AAPG 2026 – PRC Collaborative project**
Funding Agency: ANR
Title: Volumetric Integrated Spatial transcriptomics Unraveling RETinal specification
Total amount: 250 000 €
Duration: 3 years (until 10/2026)

Name: **CENTURI Group Leader position**
Funding Agency: ANR
Total amount: ∼900 000 € (Note that these values are personal estimations)
Detailed Budget: 150 000 €, Personal salary: ∼375 000 €, ∼5 years salaries: ∼400 000 €
Duration: 5 years (until 06/2026)

Name: **Research Grants – Program**
Funding Agency: Human Frontier Science Program
Title: Cellular and molecular basis of bilaterian symmetry
Total amount: 400 000 $ (shared between 3 groups)
Duration: 3 years (until 10/2026)

Name: **Interdisciplinarity 2021**
Funding Agency: A*MIDEX – Marseille
Title: Neural representation of social information in hippocampal ventral CA1 of normal versus autistic mice
Total amount: 150 000 € (shared between 3 groups)
Duration: 2 years (until 01/2025)

Name: **Interdisciplinarity 2021**
Funding Agency: A*MIDEX – Marseille
Title: Cellular interactions visualized in 3D within whole tumors
Total amount: 100 000 € (shared between 2 groups)
Duration: 2 years (until 01/2025)

AWARDS & FELLOWSHIPS

Awards

Young Scientist Award, 2021 Health & Biology Montpellier
Best talk award, 2020 Berlin Post-doc Day
Best poster, 2020 LSFM conference
Best early-career talk, 2019 SFBD/SBCF conference

Fellowships

Humboldt renewed research stay: 02/23 – 05/23
Humboldt Research Fellowship for Postdoc: 10/20 – 09/21
4th year Ph.D fellowship, Fondation pour la Recherche Médicale: 09/14 – 08/15

MENTORING AND PH.D. COMMITTEES

Mentoring

Sendra M. (Postdoc): 2025 – present
Dynar M. (Postdoc): 2025 – present
Liaskas I. (Ph.D.): 2025 – present
Mazzerbo C. (Ph.D.): 2025 – present
Song S. (Postdoc): 2025 – 2026 (Currently CR-CN INSERM)
Loof G. (Postdoc): 2022 – 2024 (Currently Software designer)
Gros A. (Ph.D.): 2021 – 2025 (Currently Postdoc)
Internships: 8
Ph.D. committees: 5

CONFERENCE ACTIVITY

Invited Speaker

Imaging Mouse Development Conference, Janelia Research Campus, US, 09/26
Institut Curie, BDD, Paris, 04/26
Microscience Microscopy Congress, Manchester, UK, 07/25
SFDB Shaping Life 3, Cassis, France: 06/25
Stem Cell Symposium, Vienna, Austria: 03/25
Spatial Biology Europe 2024, Basel, Switzerland: 10/24
Virtual Gastrulation Zoom Talks, online: 02/24
Imaging Life: The Future, Montpellier, FR: 10/23
Curie Institute, Paris, FR: 05/23
Pasteur Institute, Paris, FR: 05/23
Mechanobiology Institute, Singapore: 02/23
Dev Mech Seminar Series, online: 02/23
Centre for Translational Stem Cell Biology, Hong Kong: 12/22
Max Planck Institute for Molecular Genetics, Berlin, FR: 04/22
Institut de Neurosciences de la Timone, Marseille, FR: 02/22
University of Southampton, Southampton, UK: 11/21
EMBL Conference – Visualizing Biological Data, Virtual: 03/21
Max Planck Institute for Molecular Genetics, Berlin, DE: 02/20
Max Delbrueck Centre for Molecular medicine, Berlin, DE: 07/19
Berlin Institute of Health, Berlin, DE: 07/19

Selected Speaker

Emerging Concepts in Cell & Developmental Biology Conference, Aarhus, DK: 09/22
Fom cells to embryo Conference, Paris, FR: 11/20
Mechanics in Developmental Biology, Paris, FR: 06/20
NEUBIAS Conference, Bordeaux, FR: 03/20
Developmental and Cell Biology of the future, Paris, France: 04/19

TEACHING

Teaching coordinator

Turing Centre for living systems (2021 – present)
Yearly introduction to Python programming language to Ph.D. students and postdocs

Teaching unit (UE) coordinator

Aix-Marseille University, Marseille, France (2023 – present)
Master 2 Computation and Mathematics for Biology: The biology, mathematics and computer science of Turing Patterns

Aix-Marseille University, Marseille, France (2023 – present)
Master 2 Computation and Mathematics for Biology: Advanced Programming

Aix-Marseille University, Marseille, France (2021 – 2023)
Master 1 Computation and Mathematics for Biology: Professional perspectives for biological systems modelling

Teaching assistant

Kavli Institute for Theoretical Physics, Santa-Barbara, CA, USA (2016)
Advanced School of Quantitative Biology – Mechanics and Mechanisms of Morphogenesis

École Normale Supérieure de Paris, Paris, France (2014 & 2015)
Yearly Interdisciplinary Spring School on Plant and Animal Morphogenesis

Nîmes University, Nîmes, France (2012 – 2014)
Practical courses: Programming, Graph theory, Data analysis – Bachelors 1 to 3 (Ph.D. monitorat)

SCIENCE ORGANIZATION

Committees	Jury president for the 2023 CENTURI Ph.D call Member of the Turing Centre for living system Research Committee: 2021 – now Jury member for the 2020 Max-Delbrueck Centre Ph.D call
Organization	5^{th} CenTuri Hackathon for quantitative biology, Marseille, FR: 2026 4^{th} CenTuri Hackathon for quantitative biology, Marseille, FR: 2025 3^{rd} CenTuri Hackathon for quantitative biology, Marseille, FR: 2024 2^{nd} CenTuri Hackathon for quantitative biology, Marseille, FR: 2023 1^{st} CenTuri Hackathon for quantitative biology, Marseille, FR: 2022 3^{rd} Franco-Japanese Dev Bio Meeting (SFBD-JSDB), Strasbourg, FR: 2022 22^{nd} 2-day Ph.D. Symposium of CRBM, Montpellier, FR, 2013

REVIEWER

Ph.D. thesis reviewed: 1
Journals: Bioinformatics; Briefings in Bioinformatics; Cell Reports; Development; Developmental Biology; Developmental Cell; eLife; Frontiers in Bioinformatics; IEEE – Transactions on Medical Imaging; Nature Genetics; Nature communications biology

PRESS HIGHLIGHTS AND SCIENCE OUTREACH

General press	
The New-York Times	This Ancient Sea Creature Builds Its Body With a Whisper, Not a Scream
Le Monde (fr)	Cris et chuchotements chez les cellules embryonnaires
Pour la Science (fr)	Pourquoi l'embryon se développe toujours de la même façon
Wired	Watch Mouse Embryos Develop Under This 4-D Microscope
The New-York Times	Watch This Blob of Cells Become an Embryo in High-Resolution

Outreach	
Large audience video	Cries or whispers between cells – science popularization animation

Specialized press	
The Biologist	Spatialomics: Life in 3D
Science	The 200-year effort to see the embryo
Nature methods	Mapping mouse development
Science	This is the most detailed view ever of a developing mouse

Updated August 25, 2026

PUBLICATIONS

Profiles: **Google Scholar - publons**

2026 - Sendra M., Bolondi A., Guignard L.† sc3D: A Comprehensive Tool for 3D Spatial Transcriptomic Analysis. Bio-protocol 16(4): e5607. 10.21769/BioProtoc.5607

2025 - Gros A., Vanaret J., Dunsing-Eichenauer V., Rostan A., Roudot P., Lenne P.-F., Guignard L.†, Tlili S.† A quantitative pipeline for whole-mount deep imaging and multiscale analysis of gastruloids. eLife 14:RP107154, 10.7554/eLife.107154.1

- Majnik J., Mantez M., Zangila S., Bugeon S., Guignard L., Platel J.-C., Cossart R. Longitudinal tracking of neuronal activity from the same cells in the developing brain using Track2p. eLife 14:RP107540, 10.7554/eLife.107540.1

- Dunsing-Eichenauer V., Hummert J., Chardès C., Schönau T., Guignard L., Galland R., Grenci G., Tillmann M., Koberling F., Nock C., Sibarita J.-B., Viasnoff V., Antolovic I. M., Erdmann R., Lenne P.-F. Fast volumetric fluorescence lifetime imaging of multicellular systems using single-objective light-sheet microscopy. Commun Biol 8, 1785

- Sendra M., McDole K., Jimenez-Carretero D., de Dios Hourcade J., Temiño S., Raiola M., Guignard L., Keller P.J., Sánchez-Cabo F., Domínguez J.N., Torresa M. Myocardium and endocardium of the early mammalian heart tube arise from independent multipotent lineages specified at the primitive streak. Developmental Cell, 10.1016/j.devcel.2025.05.002

2023 - Sampath Kuma A., Tian L., Bolondi A., Stickel R., Hernández A., Walther M., Haut L., Murray E., Wittler L., Kretzmer H., Timmermann B., Elkabetz Y., Macosko E.†, Guignard L.†, Chen F.†, Meissner A.†, Spatiotemporal transcriptomic maps of whole mouse embryos at the onset of organogenesis. Nature Genetics, 55, 1176–1185

- Malin-Mayor C., Hirsch P., Guignard L., McDole K., Wan Y., Lemon W., Keller P., Preibisch S., Funke J. Automated reconstruction of whole-embryo cell lineages by learning from sparse annotations, Nature Biotechnology, 41, 44–49 (2023)

2022 - Hirsch P., Epstein L., Guignard L.†, Mathematical and bioinformatical tools for cell tracking, book chapter in M. Schnoor (Ed.), Cell Movement in Health and Disease, Elsevier, Pages 341-361

2020 + Veenvliet J., Bolondi B., Kretzmer H., Haut L., Scholze-Wittler M., Schifferl D., Koch F., Guignard L., Kumar A., Pustet M., Heimann S., Buschow R., Wittler L., Timmermann B., Meissner A., Herrmann B., Mouse embryonic stem cells self-organize into trunk-like structures with neural tube and somites, Science,370(6522):aba4937
+ Guignard L.*, Fiuza U.-M.*, Leggio B., Laussu J., Faure E., K., Michelin G., Biasuz K., Hufnagel L., Malandain G., Godin C., Lemaire P., Contact area-dependent cell communication and the morphological invariance of ascidian embryogenesis, Science, 369(6500):eaar5663

2019 - Madgwick A., Silvia Magri M., Dantec C., Gailly D., Fiuza U.-M., Guignard L., Hettinger S., Gomez-Skarmeta J.-L., Lemaire P., Evolution of embryonic cis-regulatory landscapes between divergent Phallusia and Ciona ascidians, Developmental Biology, 448(2):71 - 87

2018 + McDole K.†, Guignard L.†, Amat F., Berger A., Malandain G., Royer L., Turaga S., Branson K., Keller P.†, In Toto Imaging and Reconstruction of Post-Implantation Mouse Development at the Single-Cell Level, Cell, 175(3):859-876
- Wolff C., Tinevez J.-Y., Pietzsch T., Stamataki E., Harich B., Guignard L., Preibisch S., Shorte S., Keller P., Tomancak P., Pavlopoulos A., Multi-view light-sheet imaging and tracking with the MaMuT software reveals the cell lineage of a direct developing arthropod limb, eLife, 7. pii:e34410

2015 - Michelin G., Guignard L., Fiuza U.-M., Lemaire P., Godin C., Malandain G., Cell pairings for ascidian embryo registration, IEEE, 12th ISBI:298-301
- Marza E., Taouji S., Barroso K., Raymond A.-A., Guignard L., Bonneu M., Pallares-Lupon N., Dupuy J.-W., Fernandez-Zapico M., Rosenbaum J., Palladino F., Dupuy D., Chevet E., Genome-wide screen identifies a novel p97/CDC-48-dependent pathway regulating ER-stress-induced gene transcription, EMBO Reports, 16(3):332-40

2014 - Guignard L.*, Godin C., Fiuza U.-M., Hufnagel L., Lemaire P., Malandain G., Spatio-temporal registration of embryo images, IEEE, 11th ISBI:778-781
- Michelin G., Guignard L., Fiuza U.-M., Malandain G., Embryo cell membranes reconstruction by tensor voting, IEEE, 11th ISBI:1259-1262

∗: first author, †: corresponding author, +: Featured in Faculty Opinions

SOFTWARE

Profiles: **Personal GitHub** - **Lab GitHub**. Except when mentioned otherwise, I solely developed the softwares. (order by total number of downloads when possible otherwise alphabetically)

- **napari-sc3d-viewer** downloads 800k (https://github.com/GuignardLab/napari-sc3D-viewer):
A light and dynamic Python viewer for 3D spatial transcriptomics implemented as a napari plugin.

- **sc3D** downloads 748k (https://github.com/GuignardLab/sc3D):
A Python library to handle 3D spatial transcriptomics

- **Registration tools** downloads 105k (https://github.com/GuignardLab/registration-tools):
A Python library based on an already existing C++ library. The registration-tools library allows users to co-register images together either in space or in time. The registrations can have multiple degrees of freedom (rigid, affine, non-linear).

- **LineageTree** downloads 37k (https://github.com/leoguignard/LineageTree):
A Python dedicated and efficient library to read, store and analyze cell lineage trees.

- **ASTEC** (https://gitlab.inria.fr/astec/astec):
A Python and C++ computer vision software to detect cells, reconstruct their shapes and track them from 3D recordings of membranes from fluorescence microscopy. This library was developed together with G. Malandain.

- **Cell-cell interaction model** (https://github.com/leoguignard/cell-cell-interaction-model):
A Python library for the cell-cell signalling model developed in Guignard, Fiuza et al. Science 2020. The model itself was developed in collaboration with B. Leggio, C. Godin and P. Lemaire

- **IO** (https://github.com/GuignardLab/IO):
A generic Python library to read and write images.

- **napari-3d-registration** (https://github.com/GuignardLab/napari-3D-registration):
A graphical user interface in Python for image registration tools. It is implemented as a napari plugin.

- **SVF** (https://github.com/leoguignard/SVF):
A Python library to compute statistical vector flows from cell trackings.

- **TARDIS** (https://github.com/leoguignard/TARDIS):
A Python library to co-register dynamic point clouds in space and time provided manual fiducials.

- **TLS-morpho** (https://github.com/leoguignard/TLS-morpho):
A Python library to quantify the morphology of Trunk-Like-Structures organoids.